ON THE RIEMANN ZETA FUNCTION AT EVEN INTEGERS

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Introduction

The Basel Problem

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

is a very famous result in Mathematics, known even to high school students. However, what is more mysterious is the underlying function behind this result, the Riemann Zeta Function, defined by the infinite series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

In this paper, I outline a proof of the identity

$$2\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{(2m)!} B_{2m},$$

where B_k denotes the k-th Bernoulli number. We do this by applying the Poisson Summation Formula and exploiting absolute convergence to derive identities involving infinite series, which will be used in the main proof in the last section. I do not claim any originality for the identities considered; see [SS03].

As always, suggestions, queries, and comments for expositional improvement will be graciously received at my e-mail:

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1. First Identity

Theorem. For $t > 0$, we have

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \sum_{n=-\infty}^{\infty} e^{-2\pi t|n|}. \tag{1}$$

Proof: Consider the function $f(x) = \frac{1}{\pi}\frac{t}{x^2+t^2}$, for $t > 0$. f is clearly of moderate decrease. Calculations show that the Fourier transform is given by $\hat{f}(\xi) = e^{-2\pi t|\xi|}$, so we may invoke the Poisson Summation Formula[1] to obtain

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \sum_{n=-\infty}^{\infty} e^{-2\pi t|n|},$$

as required. ■

2. More Identities

Theorem. When $0 < t < 1$, we have the identities

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \frac{1}{\pi t} + \frac{2}{\pi}\sum_{m=1}^{\infty}(-1)^{m+1}\zeta(2m)t^{2m-1}, \tag{2}$$

$$\sum_{n=-\infty}^{\infty} e^{-2\pi t|n|} = \frac{2}{1-e^{-2\pi t}} - 1, \tag{3}$$

where ζ denotes the Riemann Zeta Function.

Proof: We begin with the first identity. Observe that [2]

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \frac{1}{\pi t} + \frac{1}{\pi}\sum_{n\neq 0}\frac{t}{t^2+n^2} = \frac{1}{\pi t} + \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{t}{t^2+k^2}.$$

Note that for $0 < t < 1$, the Maclaurin Series of $\frac{t}{t^2+m^2}$ is

$$\frac{t}{t^2+k^2} = \sum_{m=1}^{\infty}(-1)^{m+1}\frac{t^{2m-1}}{k^{2m}}.$$

It follows that

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \frac{1}{\pi t} + \frac{2}{\pi}\sum_{k=1}^{\infty}\sum_{m=1}^{\infty}(-1)^{m+1}\frac{t^{2m-1}}{k^{2m}}.$$

The absolute convergence of the series above permits us to interchange sums:

[1]For functions of moderate decrease, we have $\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \hat{f}(n)$.

[2]I changed the index for cleanliness.

$$\begin{aligned}
\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} &= \frac{1}{\pi t}+\frac{2}{\pi}\sum_{m=1}^{\infty}\sum_{k=1}^{\infty}(-1)^{m+1}\frac{t^{2m-1}}{k^{2m}}\\
&= \frac{1}{\pi t}+\frac{2}{\pi}\sum_{m=1}^{\infty}(-1)^{m+1}t^{2m-1}\sum_{k=1}^{\infty}\frac{1}{k^{2m}}\\
&= \frac{1}{\pi t}+\frac{2}{\pi}\sum_{m=1}^{\infty}(-1)^{m+1}\zeta(2m)t^{2m-1},
\end{aligned}$$

proving Equation (2).

The proof of Equation 3 is a trivial application of the geometric series formula; observe that

$$\begin{aligned}
\sum_{n=-\infty}^{\infty}e^{-2\pi t|n|} &= 1+2\sum_{n=1}^{\infty}e^{-2\pi tn}\\
&= 1+\frac{2e^{-2\pi t}}{1-e^{-2\pi t}}\\
&= \frac{2}{1-e^{-2\pi t}}-1,
\end{aligned}$$

as desired. ■

3. Main Result

We are now ready to prove the identity discussed in the Introduction.

Theorem. Let B_k denote k-th Bernoulli number. Then we have the formula

$$2\zeta(2m) = (-1)^{m+1}\frac{(2\pi)^{2m}}{(2m)!}B_{2m}. \tag{4}$$

Proof: We recall the identity

$$\frac{z}{e^z-1} = 1-\frac{z}{2}+\sum_{m=1}^{\infty}\frac{B_{2m}}{(2m)!}z^{2m}.$$

We substitute $z=-2\pi t$ with $0<t<1$ into the equation above, to obtain

$$\begin{aligned}
\sum_{m=1}^{\infty}\frac{B_{2m}(2\pi)^{2m}}{(2m)!}t^{2m} &= -\frac{2\pi t}{e^{-2\pi t}-1}-1-\pi t = -1-\pi t\left(\frac{2}{e^{-2\pi t}-1}+1\right)\\
&= -1+\pi t\left(\frac{2}{1-e^{-2\pi t}}-1\right)\\
&= -1+\pi t\sum_{n=-\infty}^{\infty}e^{-2\pi t|n|}.
\end{aligned}$$

From Equation 1,

$$\begin{aligned}\sum_{m=1}^{\infty}\frac{B_{2m}(2\pi)^{2m}}{(2m)!}t^{2m} &= -1+\pi t\left(\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2}\right)\\ &= -1+\pi t\left(\frac{1}{\pi t}+\frac{2}{\pi}\sum_{m=1}^{\infty}(-1)^{m+1}\zeta(2m)t^{2m-1}\right)\\ &= \sum_{m=1}^{\infty}2(-1)^{m+1}\zeta(2m)t^{2m},\end{aligned}$$

where the second equality holds by Equation 2.

By the uniqueness property of power series[3], we have that

$$\frac{B_{2m}(2\pi)^{2m}}{(2m)!} = 2(-1)^{m+1}\zeta(2m),$$

which is to say

$$2\zeta(2m) = (-1)^{m+1}\frac{(2\pi)^{2m}}{(2m)!}B_{2m},$$

as required. ■

Substituting $m = 2$, we obtain the result of the Basel Problem.

References

[Rud76] Walter Rudin. *Principles of Mathematical Analysis.* 3rd. New York: McGraw-Hill, 1976.

[SS03] Elias M. Stein and Rami Shakarchi. *Fourier Analysis: An Introduction.* Vol. 1. Princeton Lectures in Analysis. Princeton University Press, 2003.

[3]See Theorem 8.5 in [Rud76]